

## Quiz – 5

Time Allowed: 30

Mins Total Marks: 20

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### The "AuthKey" Access Manager

**Instructions:** You are tasked with implementing a C++ class that manages temporary access keys for a secure system.

**Part 1:** Complete the AuthKey class definition below. Fill in the code exactly where the comments `/* ... */` instruct you to.

**Part 2:** Write the main() function. Follow the high-level instructions within the main() comments to implement the required logic.

**Part 3:** In the "Predicted Output" section, write the *exact* text that will be printed to the console when your completed main() function is executed.

```
#include <iostream>
```

```
#include <string>
```

```
class AuthKey {
```

```
private:
```

```
    string user;      // The name of the user assigned to this key
```

```
    int pinCode;     // The key's unique 4-digit PIN code
```

```
    string status;   // The current status of the key (e.g., "Active", "Locked")
```

```
    int attemptsLeft; // How many login attempts are left before it locks
```

```
public:
```

```
    // --- Constructor ---
```

```
    // 1. Initialize 'user' and 'pinCode' from parameters. Set 'status' to "Active" and 'attemptsLeft' to 3. Print "Key created for [user]."
```

```
    AuthKey(string userName, int pin) {
```

```
/* YOUR CODE HERE */

}

// --- Destructor
// Print "Key for [user] destroyed."
~AuthKey() {
    /* YOUR CODE HERE */
}

// --- Getter
string getUser() {
    /* YOUR CODE HERE */
}

string getStatus() {
    /* YOUR CODE HERE */
}

// Resets the PIN code.
// 1. The 'newPin' must be a valid 4-digit number.
// 2. If valid: update 'pinCode', reset 'attemptsLeft' to 3, set 'status' to "Active", and print "PIN
reset successful."
// 3. If invalid, print "Error: Invalid PIN."
```

```
void resetPin(int newPin) {
```

```
    /* YOUR CODE HERE */
```

```
}
```

```
// Attempts to log in using the key.
```

```
// 1. If 'status' is "Locked", print message
```

```
// 2. If 'attemptedPin' is correct: Print successful message - Reset 'attemptsLeft' to 3.
```

```
// 3. If 'attemptedPin' is incorrect: Decrement 'attemptsLeft'. Print message
```

```
// If 'attemptsLeft' reaches 0, set lock the user.
```

```
void login(int attemptedPin) {
```

```
    /* YOUR CODE HERE */
```

```
}
```

```
};
```

```
int main() {
```

```
// Create an AuthKey for user "Admin" with PIN 1234.
```

```
// Attempt to log in with PIN 1111, then with PIN 1234.
```

```
// Print "--- Starting inner scope ---"
```

```
{
```

```
    // Create 'guestKey' for user "Guest" with PIN 9999.
```

```
    // Attempt three logins with wrong PIN 1234.
```

```
    // Print status after each relevant step.
```

```
    // Attempt correct PIN 9999.
```

```
    /* YOUR CODE HERE */
```

```
} // End of inner scope
```

```
// Print "--- Resuming main scope ---"  
  
// Test PIN reset (invalid first, then valid),  
  
// attempt login with old PIN, and display final status.  
  
/* YOUR CODE HERE */
```

```
    return 0;  
}
```

Write your output here.